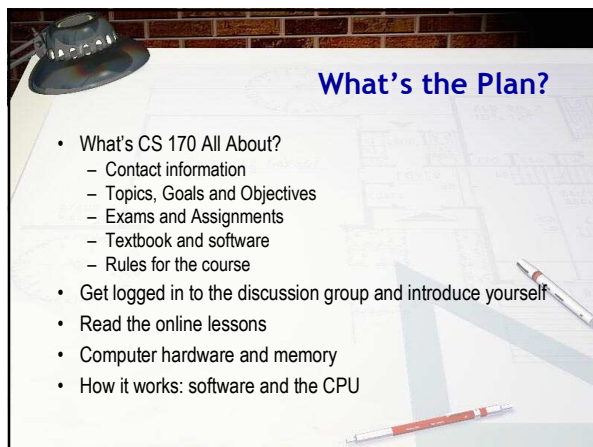
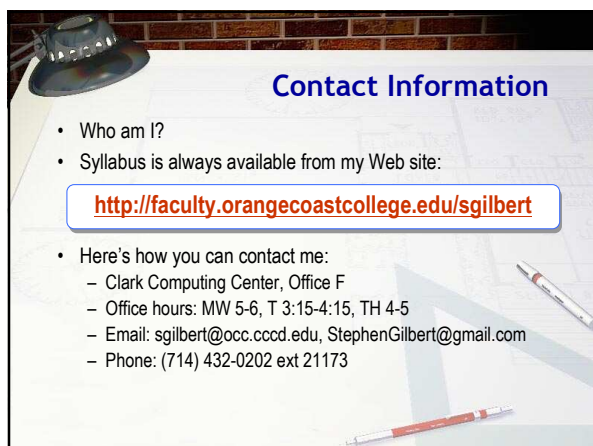
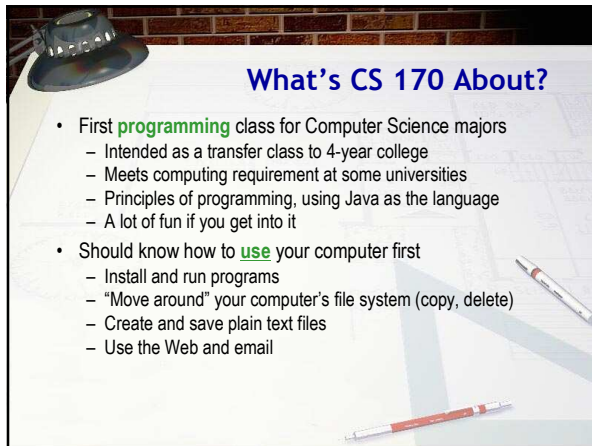




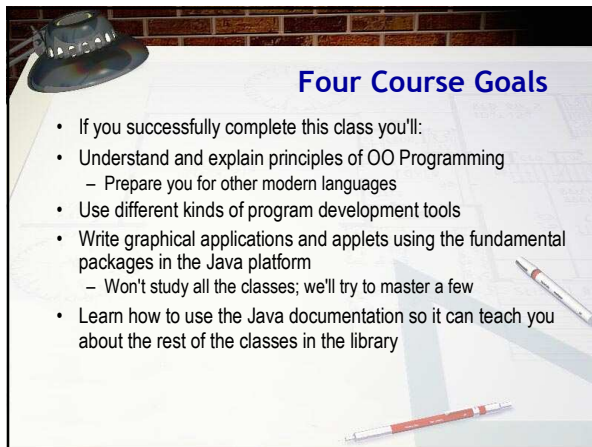






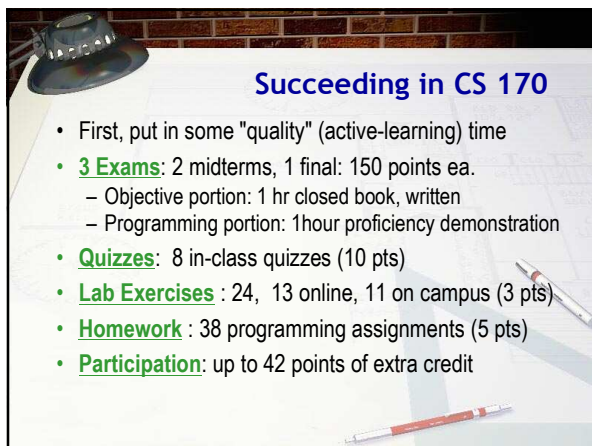
What's CS 170 About?

- First **programming** class for Computer Science majors
 - Intended as a transfer class to 4-year college
 - Meets computing requirement at some universities
 - Principles of programming, using Java as the language
 - A lot of fun if you get into it
- Should know how to **use** your computer first
 - Install and run programs
 - "Move around" your computer's file system (copy, delete)
 - Create and save plain text files
 - Use the Web and email



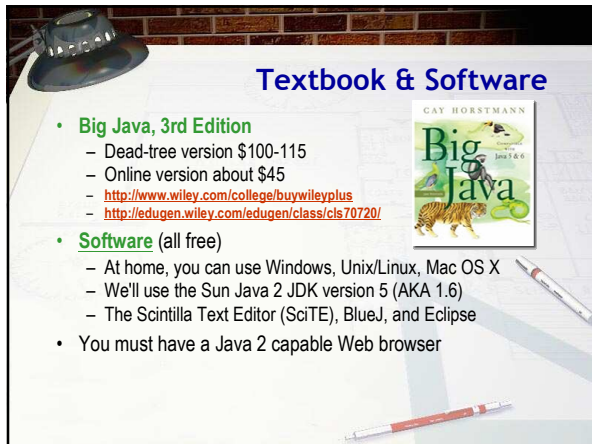
Four Course Goals

- If you successfully complete this class you'll:
- Understand and explain principles of OO Programming
 - Prepare you for other modern languages
- Use different kinds of program development tools
- Write graphical applications and applets using the fundamental packages in the Java platform
 - Won't study all the classes; we'll try to master a few
- Learn how to use the Java documentation so it can teach you about the rest of the classes in the library



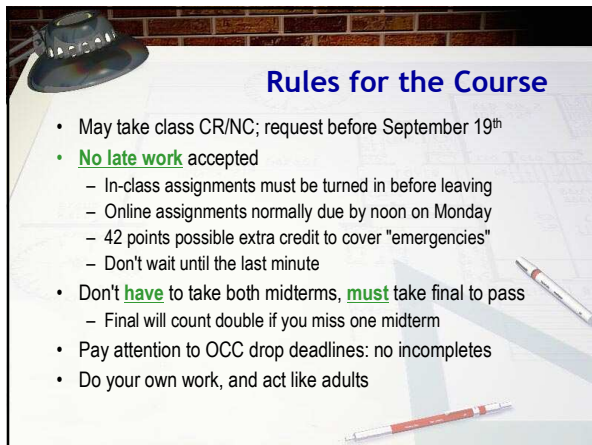
Succeeding in CS 170

- First, put in some "quality" (active-learning) time
- **3 Exams**: 2 midterms, 1 final: 150 points ea.
 - Objective portion: 1 hr closed book, written
 - Programming portion: 1hour proficiency demonstration
- **Quizzes**: 8 in-class quizzes (10 pts)
- **Lab Exercises**: 24, 13 online, 11 on campus (3 pts)
- **Homework**: 38 programming assignments (5 pts)
- **Participation**: up to 42 points of extra credit



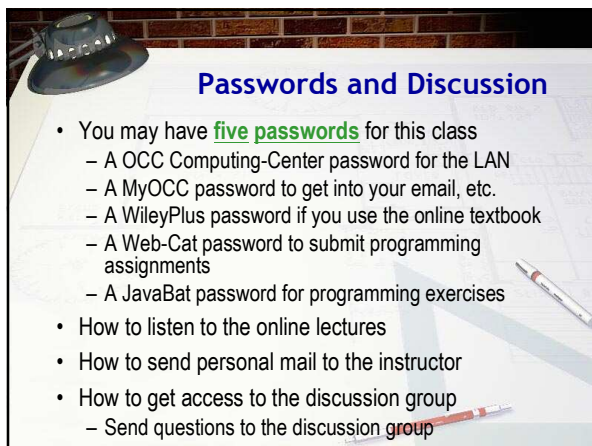
Textbook & Software

- **Big Java, 3rd Edition**
 - Dead-tree version \$100-115
 - Online version about \$45
 - <http://www.wiley.com/college/buywileyplus>
 - <http://edugen.wiley.com/edugen/class/cis70720/>
- **Software** (all free)
 - At home, you can use Windows, Unix/Linux, Mac OS X
 - We'll use the Sun Java 2 JDK version 5 (AKA 1.6)
 - The Scintilla Text Editor (SciTE), BlueJ, and Eclipse
- You must have a Java 2 capable Web browser



Rules for the Course

- May take class CR/NC; request before September 19th
- **No late work** accepted
 - In-class assignments must be turned in before leaving
 - Online assignments normally due by noon on Monday
 - 42 points possible extra credit to cover "emergencies"
 - Don't wait until the last minute
- Don't **have** to take both midterms, **must** take final to pass
 - Final will count double if you miss one midterm
- Pay attention to OCC drop deadlines: no incompletes
- Do your own work, and act like adults

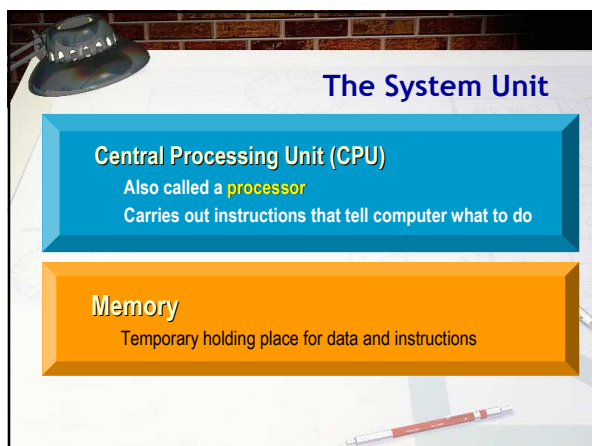


Passwords and Discussion

- You may have **five passwords** for this class
 - A OCC Computing-Center password for the LAN
 - A MyOCC password to get into your email, etc.
 - A WileyPlus password if you use the online textbook
 - A Web-Cat password to submit programming assignments
 - A JavaBat password for programming exercises
- How to listen to the online lectures
- How to send personal mail to the instructor
- How to get access to the discussion group
 - Send questions to the discussion group







Representing Data

- The CPU and memory are digital (transistor-like) devices
 - Transistors is always on or off
 - Can treat state as a **binary number**
 - Single binary digit is called a **bit**
- Today's computers are usually **byte-orient**
 - Byte is a group of 8 bits
 - Represent 256 values
 - Numbers, characters, instructions
- Each byte in memory is stored in a unique location, called its **address**

Introducing Debug

- Let's use **debug** to examine memory and the CPU
- With debug you can:
 - View and change the contents of memory
 - View, create, and modify **any type** of file
 - Convert assembly language programs into executable code
 - Convert machine language files into assembly source files
 - Examine any and all of the CPU registers
 - Trace the execution of a program, step by step
- We'll examine several different aspects of computer memory

Getting Started

- First we'll examine a text file loaded into memory
 - Press Win+R to open Run, type **cmd**, press **ENTER**
 - Type **Notepad myfile.txt** and press **ENTER**
 - Type **"Sam was a man 123"** like this and save the file
 - Type **DIR** in command window, and locate your file

Starting Debug

- At the command prompt type: `debug myfile.txt`
 - Type `d` to dump or display memory, then press **ENTER**

segment:offset address format

16 bytes

increments by 10H = 16 bytes

ASCII codes stored in file

ASCII translation

Explain Yourself

- Use OpenOffice or Word to create a new document
- Save the file as LastFirst_ic01.doc
 - Replace LastFirst with your actual name
- Put your name on the top of the sheet
- Select the debug window, press ALT+Print Screen
- Select your document and paste the picture
- Below the picture explain:
 - Why does each byte require two numbers to represent it?
 - Locate the 123 you typed: what bytes are stored in memory?
 - Why isn't the byte value for the number 1, 1?
- Save your file

Video Memory

- Let's take a look at text-mode video memory
 - Type `-d B800:00A0` and press **ENTER**
 - Address of second line on screen
 - First byte is ASCII character, second is the attribute

segment:offset address format

16 bytes

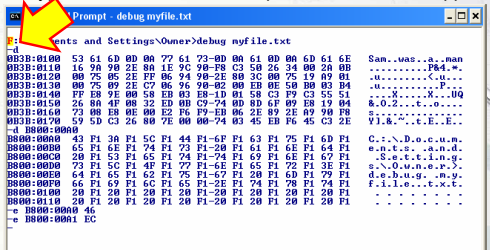
increments by 10H = 16 bytes

ASCII codes stored in file

ASCII translation

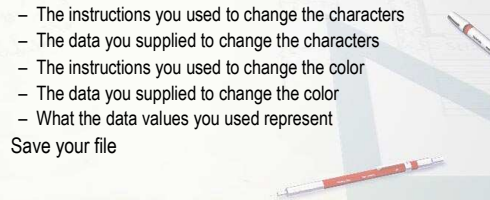
Changing Video Memory

- Type `-e B800:00A0 46` and press **ENTER**
- ASCII value on second line changes from C to F
- Type `-e B800:00A1 EC` and press Enter. What happens?



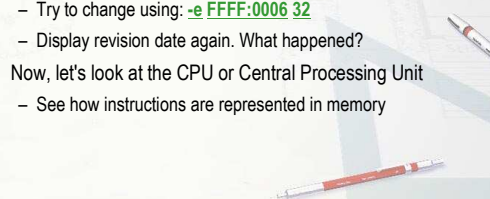
Explain Yourself

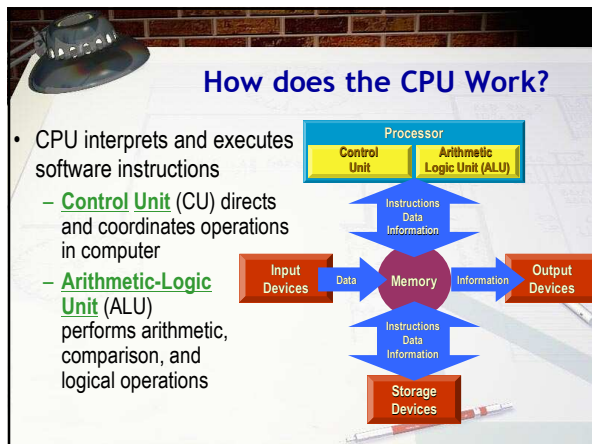
- Change the first two characters on the second line so that your initials are displayed (start and stop debug if scrolling)
- Change the color of those two characters to blue on white
- Snap a picture and paste it in your document
- Below the picture explain:
 - The instructions you used to change the characters
 - The data you supplied to change the characters
 - The instructions you used to change the color
 - The data you supplied to change the color
 - What the data values you used represent
- Save your file

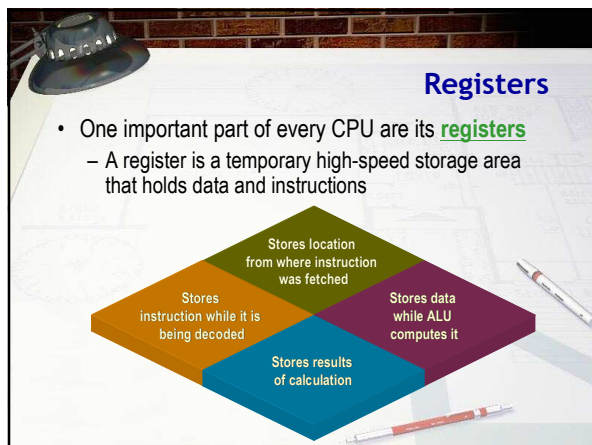


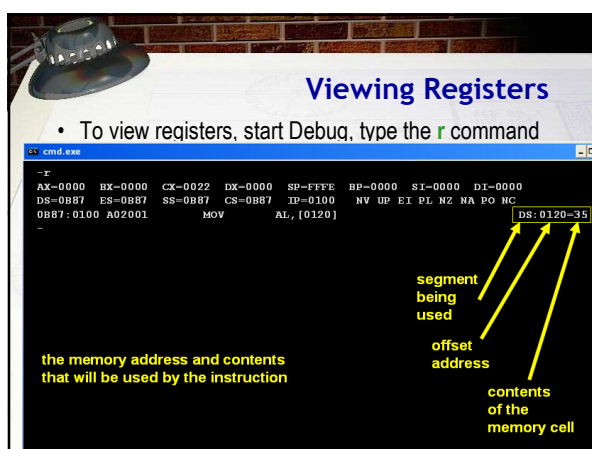
Read-Only Memory

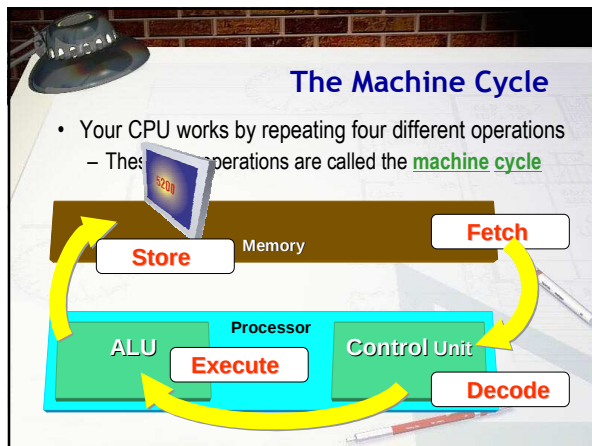
- ROM BIOS:** permanent code that boots your computer
 - ROM BIOS starts at location **FFFF**
 - BIOS revision date is at **offset 0005**
 - Display revision date by typing: `-d FFFF:0005`
 - Try to change using: `-e FFFF:0006 32`
 - Display revision date again. What happened?
- Now, let's look at the CPU or Central Processing Unit
 - See how instructions are represented in memory











Fetch and Decode

- Let's look at instruction fetched when you power up
 - Computer **hardwired** to fetch instruction at **FFFF:0000**

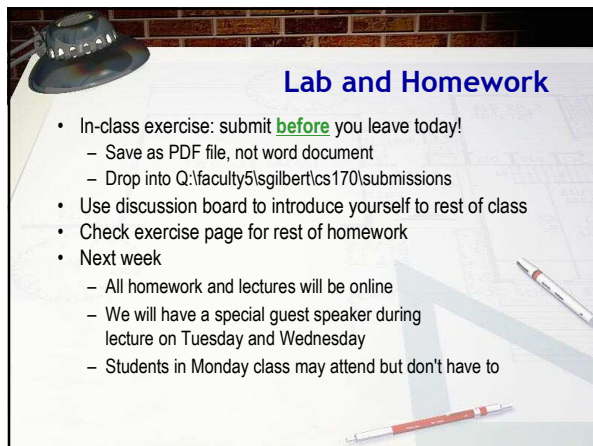
```

cmd.exe
~D FFFF:0
FFFF:0000 EA5BE00F0 JMP F000:E05B
FFFF:0005 3035 XOR [DI],DH
FFFF:0007 2F DAS [BX+DI],DI
FFFF:0008 3139 XOR [BX+DI],DI
FFFF:000A 2F DAS [BP+DI],DH
FFFF:000B 3033 XOR [BP+DI],DH
FFFF:000D 00FC ADD AH,BH
FFFF:000F 0034 ADD [SI],BH
FFFF:0011 1200 ADC AX,[BX+SI]
FFFF:0013 0000 ADD [BX+SI],AL
FFFF:0015 0000 ADD [BX+SI],AL
FFFF:0017 0000 ADD [BX+SI],AL
FFFF:0019 0000 ADD [BX+SI],AL
FFFF:001B 0000 ADD [BX+SI],AL
FFFF:001D 0000 ADD [BX+SI],AL
FFFF:001F 0000 ADD [BX+SI],AL
  
```

These are not really instructions!
BIOS revision data

The Command Line

- Some things are easier from the command-line
- How do you: (type answers in ic01)
 - Open a command-line shell window?
 - Change to a different device or drive?
 - Change to a different working directory?
 - Locate files in a directory using wildcards?
 - Copy, delete, rename or move a file?
 - Create or remove a directory?
 - Run a program from the command line?
 - Change the executable search path?
 - Examine your execution environment?



Lab and Homework

- In-class exercise: submit **before** you leave today!
 - Save as PDF file, not word document
 - Drop into Q:\faculty5\sgilbert\cs170\submissions
- Use discussion board to introduce yourself to rest of class
- Check exercise page for rest of homework
- Next week
 - All homework and lectures will be online
 - We will have a special guest speaker during lecture on Tuesday and Wednesday
 - Students in Monday class may attend but don't have to
